

SOFTWARE ENGINEERING

LAB – 3

Component Modelling & Architectural Pattern Selection *Problem Statement #56 — Media, Events & Community* *Peer Skill Exchange & Mentorship Network*

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SECTION – B

Architectural Justification

Architecture Selection — Layered Architecture

We chose Layered Architecture for the Peer Skill Exchange & Mentorship Network. The system is organized into presentation, business, integration, and data responsibilities. The User Interface component handles interaction with Learner Members and Skill Mentors, while the business components manage skill exchange, sessions, time credits, and feedback.

Reason 1 — Separation of Concerns

Layered Architecture separates user interaction from business logic and data management. This makes the system easier to understand, maintain, and modify. For example, changes to the user interface can be made without changing the time-credit management functionality.

Reason 2 — Maintainability and Organization

The system contains related functions such as searching and maintaining skills, booking skill-exchange sessions, confirming session completion, managing time credits, and handling feedback. Separating these responsibilities into components gives each part a clear role and supports incremental development and maintenance.

Security Advantage

The architecture separates access to member accounts, session information, time credits, and feedback data from the presentation layer. Access to these resources can therefore be controlled through the appropriate business components, supporting protection against unauthorized access.

Performance Benefit

The components provide direct communication paths for common operations such as session management and time-credit updates. This avoids unnecessary processing through unrelated functionality and keeps the flow of frequent operations straightforward.

Components and Interfaces

Components	Interfaces / Responsibility
User Interface Component	Skill/Session API — user interaction with skill and session functions
Skill Exchange & Session Management	Time Credit API; Feedback API; Calendar Sync API; Session Data API
Time Credit Management Component	Credit Data API — read/write credit records
Feedback & Rating Management Component	Feedback Data API — store/retrieve feedback
Calendar Integration Component	Calendar Sync API — synchronize scheduled sessions
Database Component	Stores members, skills, sessions, time credits, and feedback